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CLINICAL RESEARCH ARTICLE

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Serum thyroid-stimulating hormone is an independent risk factor of recurrent Guillain-Barré syndrome

Shi Wang MS[1] | Zhiya Luo MS[2] | Tao Peng PhD[1]

[1]Department of Neurology, The First Affiliated Hospital of Zhengzhou University, Zhengzhou, China

[2]Department of Otolaryngology, The First Affiliated Hospital of Dalian Medical University, Dalian, China

Correspondence
Tao Peng, PhD, Department of Neurology, The First Affiliated Hospital of Zhengzhou University, Zhengzhou 450052, China.
Email: pengtao2022@126.com

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Abstract

Introduction/Aims: Guillain-Barré syndrome (GBS) is generally considered to be monophasic, but some patients have recurrences. The purpose of this study was to clarify the possible link between thyroid parameters and recurrent GBS (RGBS) patients in China.

Methods: In this retrospective study we enrolled patients who were admitted to the Department of Neurology of The First Affiliated Hospital of Zhengzhou University from 2014 to 2020 and fulfilled the diagnostic criteria of GBS or Miller Fisher syndrome. We evaluated clinical characteristics; cerebrospinal fluid parameters; and serum levels of thyroid-stimulating hormone (TSH), free thyroxine, and free triiodothyronine in 320 individuals, including 302 with monophasic GBS and 18 with recurrent GBS.

Results: Serum levels of TSH in monophasic GBS patients were significantly lower than those in RGBS patients ($P < .001$), whereas FT3 levels were higher in the monophasic GBS group ($P = .022$). Age at onset, incidence of antecedent illness, time from onset to nadir, proportion with acute inflammatory demyelinating polyradiculoneuropathy, and Hughes Functional Grading Scale score at nadir were statistically significant between monophasic GBS patients and RGBS patients ($P < .05$). The multivariate regression analysis revealed that antecedent illness, AIDP, and high TSH were independent risk factors for RGBS. Our receiver-operating characteristic curve analysis showed that the risk of recurrence in GBS patients increases when the TSH concentration is higher than 3.87 μIU/mL.

Discussion: Our results demonstrate an association between TSH and RGBS. Oxidative stress is one of the possible interpretations for this association.

KEYWORDS
free thyroxine, free triiodothyronine, Guillain-Barré syndrome, recurrence, thyroid-stimulating hormone

Abbreviations: A-CIDP, chronic inflammatory demyelinating polyneuropathy with acute onset; AIDP, acute inflammatory demyelinating polyradiculoneuropathy; ALS, amyotrophic lateral sclerosis; AUC, area under the curve; CIDP, chronic inflammatory demyelinating polyneuropathy; CNS, central nervous system; CSF, cerebrospinal fluid; FT3, free triiodothyronine; FT4, free thyroxine; GBS, Guillain-Barré syndrome; GBS-TRF, GBS patients with treatment-related fluctuations; HFGS, Hughes Functional Grading Scale; MAP, mitogen-activated protein; MDA, malondialdehyde; MFS, Miller Fisher syndrome; RGBS, recurrent Guillain-Barré syndrome; ROS, reactive oxygen species; TH, thyroid hormone; TSH, thyroid-stimulating hormone.

1 | INTRODUCTION

Guillain-Barré syndrome (GBS) is an immune-mediated peripheral neuropathy.[1] It is generally considered to be a monophasic disease, but some patients have recurrences after long periods of remission. The frequency of recurrent Guillain-Barré syndrome (RGBS) has been

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reported at approximately 2% to 6%.[2,3] and the recurrence rate of its variant, Miller Fisher syndrome (MFS), is up to 12%.[4] Higher numbers of GBS recurrences are associated with increased disease severity and disability.[5] It is unclear why some patients have a recurrence, whether some GBS variants recur more frequently, and if predictors of recurrence exist. Early prediction of recurrence in patients with GBS would be of great significance, enabling clinicians to individualize treatment and improve outcomes. Recently, a systematic study revealed that thyroid autoantibodies may be used as biomarkers to distinguish GBS from chronic inflammatory demyelinating polyneuropathy (CIDP).[6] Therefore, the purpose of this study was to clarify the possible link between thyroid parameters and recurrent Guillain-Barré syndrome (RGBS) in China.

2 | METHODS

2.1 | Subjects

In this retrospective study we enrolled patients who were admitted to the Department of Neurology of The First Affiliated Hospital of Zhengzhou University from 2014 to 2020 and had fulfilled the diagnostic criteria of GBS and MFS.[7,8] Patients with RGBS were identified according to the following criteria: (1) at least two acute episodes with a nadir within 4 weeks, fulfilling the diagnostic criteria for GBS and MFS; and (2) an interval of at least 4 months between episodes if recovery was incomplete (GBS disability scale score ≥2) or minimum interval of at least 2 months when there was complete or nearly complete recovery (GBS disability scale score <2) after the previous episode. Participant exclusion criteria were: (1) being less than 18 years of age; (2) having GBS with treatment-related fluctuations (GBS-TRF) or chronic inflammatory demyelinating polyneuropathy with acute onset (A-CIDP), with A-CIDP being defined as CIDP with a nadir occurring in the first episode within 8 weeks of onset, and the consecutive course being chronic, as in CIDP[9]; (3) a history of thyroid disease, treated or untreated; and (4) having complications of neurological deficits that would affect disease assessment.

2.2 | 2.2 Thyroid parameters in serum

Because GBS needs to be differentiated from peripheral neuropathy caused by thyroid disease, most GBS patients in our hospital are tested for thyroid function within 48 hours after admission. A blood sample was collected in the morning within 48 hours of admission to measure free thyroxine (FT4), free triiodothyronine (FT3), and thyroid-stimulating hormone (TSH).

2.3 | Clinical characteristics data

For all included subjects, the following data were collected: age at onset; sex; antecedent illness (mainly including upper respiratory infection and diarrhea, with the time of antecedent illness, defined as 1 to 4 weeks before the onset of GBS); time from onset to nadir; Hughes Functional Grading Scale (HFGS)[10] score at nadir; sensory disturbances and cranial nerve deficits; autonomic dysfunction; mechanical ventilation; and electrophysiological findings. Electrodiagnosis of acute inflammatory demyelinating polyneuropathy (AIDP) was based on nerve conduction criteria established by Ho and colleagues.[11]

2.4 | Cerebrospinal fluid parameters

Cerebrospinal fluid (CSF) data were collected, including opening pressure, white blood cell count, protein content, and chloride.

2.5 | Statistical analysis

SPSS version 23.0 (IBM Corp, Armonk, New York) was used for data processing and analysis. Measurement data conforming to a normal distribution were expressed as mean ± standard deviation, and an independent sample *t* test was used for comparison; measurement data not conforming to a normal distribution were expressed as median and interquartile range, and the Mann-Whitney *U* test was used for comparison. Count data are expressed as ratio or percent, and were compared using the chi-square test or Fisher exact test. Multivariate logistic regression was used to analyze the factors affecting RGBS. A receiver-operating characteristic (ROC) curve was used to evaluate the effect of thyroid parameters on GBS recurrence and determine the optimal cut-off value for hormone level. A two-sided $P < .05$ was considered statistically significant for all statistical tests.

This study was approved by the ethics committee of The First Affiliated Hospital of Zhengzhou University.

3 | RESULTS

3.1 | Comparison of thyroid parameters

We found that serum levels of TSH in patients with monophasic GBS were significantly lower when compared with those of RGBS patients, but no statistically significant difference was observed for FT4 levels between monophasic GBS and RGBS. FT3 levels were significantly higher in the monophasic GBS group. Comparisons of the aforementioned indicators are shown in Table 1.

3.2 | Comparison of clinical characteristics and CSF parameters

The clinical characteristics of patients with monophasic GBS and RGBS during their first episodes are shown in Table 2. A total of 320 patients were enrolled in this retrospective study. The monophasic GBS group (302 patients) included 283 patients with typical GBS

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TABLE 1 Comparison of thyroid parameters between monophasic GBS patients and RGBS patients

Characteristics	Monophasic GBS (n = 302)	Recurrent GBS (n = 18)	*Z*	*P* value
TSH 0.34-5.60, μIU/mL	1.81 (1.03-2.75)	4.42 (3.94-5.07)	−5.682	<.001[a]
FT3 3.8-7.0, pmol/L	4.89 (4.34-5.73)	4.40 (4.08-4.79)	−2.289	.022[a]
FT4 7.9-18.4, pmol/L	12.34 (10.60-14.28)	13.90 (11.41-15.43)	−1.904	.057

Abbreviations: FT3, free triiodothyronine; FT4, free thyroxine; GBS, Guillain-Barré syndrome; RGBS, recurrent Guillain-Barré syndrome; TSH, thyroid-stimulating hormone.
Note: Data expressed as median (interquartile range).
Bold values are statistically significant.
[a]Statistically significant (*P* < .05).

TABLE 2 Comparison of characteristics between monophasic GBS patients and RGBS patients

Characteristics	Monophasic GBS (n = 302)	Recurrent GBS (n = 18)	Z/χ^2	*P* value
Male	193 (64%)	8 (44%)	2.755	.097
Age (years)	51.5 (37.0-63.0)	46.5 (37.3-51.0)	−2.012	.044[a]
Antecedent illness	143 (47%)	14 (78%)	6.293	.012[a]
Time from onset to nadir (days)	7 (4-10)	5 (4-7)	−2.069	.039[a]
Sensory disturbances	197 (65%)	13 (72%)	0.368	.544
HFGS at nadir	3 (2-4)	2 (2-3)	−2.045	.041[a]
Cranial nerve deficits	117 (39%)	7 (39%)	0.000	.990
Autonomic dysfunctions	67 (22%)	3 (17%)		.772
Mechanical ventilation	41 (14%)	1 (6%)		.486
Electrophysiology: AIDP	129 (43%)	15 (83%)	11.324	.001[a]
MFS	19 (6%)	1 (6%)		1.000
Opening pressure (mmH_2O)	150 (130-180)	170 (149-193)	−1.745	.081
CSF white blood cell count (cells/mm^3)	2 (0-4)	2 (0-2)	−1.916	.055
CSF protein content (g/L)	0.69 (0.50-1.19)	0.77 (0.54-0.97)	−0.180	.857
CSF chloride (mmol/L)	124.0 (123.0-126.7)	123.5 (123.0-127.0)	−0.181	.857

Abbreviations: AIDP, acute inflammatory demyelinating polyneuropathy; CSF, cerebrospinal fluid; GBS, Guillain-Barré syndrome; HFGS, Hughes Functional Grading Scale; MFS, Miller Fisher syndrome; RGBS, recurrent Guillain-Barré syndrome.
Note: Data expressed as number (%) or as median (interquartile range).
[a]Statistically significant (*P* < .05).

and 19 patients with MFS. The RGBS group included 18 patients, excluding those with GBS-TRF and A-CIDP. Compared with monophasic GBS patients, those with RGBS were significantly younger at age at onset and had a shorter time from onset to nadir. Furthermore, we found that antecedent illness was more common in the RGBS patients, whose HFGS at nadir was significantly lower. AIDP was significantly more common in the recurrent patients than in the monophasic GBS patients. However, cranial nerve deficits, autonomic dysfunction, mechanical ventilation, and MFS were not statistically different between the two groups. In addition, we compared CSF parameters and found them to be comparable between the two groups (Table 2).

3.3 | Multivariate logistic regression analysis of factors for RGBS patients

Indicators that were significantly different on univariate analysis were included in multivariate logistic regression analysis (Table 2). As shown in Table 3, multivariate logistic analysis revealed that antecedent illness, AIDP, and high TSH were independent risk factors for RGBS, adjusted for other risk factors.

TABLE 3 TSH and other characteristics in multivariate logistic regression analysis

Characteristics	OR	95% CI	*P* value
Age (years)	0.975	0.937-1.015	.214
Antecedent illness	5.373	1.318-21.904	.019[a]
Time from onset to nadir (days)	0.842	0.676-1.049	.126
HFGS at nadir	0.544	0.277-1.069	.077
Electrophysiology: AIDP	5.371	1.086-26.576	.039[a]
TSH (μIU/ml)	3.902	2.073-7.342	<.001[a]
FT3 (pmol/L)	0.582	0.243-1.393	.224

Abbreviations: AIDP, acute inflammatory demyelinating polyneuropathy; CI, confidence interval; FT3, free triiodothyronine; HFGS, Hughes Functional Grading Scale; OR, odds ratio; TSH, thyroid-stimulating hormone.
[a]Statistically significant (*P* < .05).

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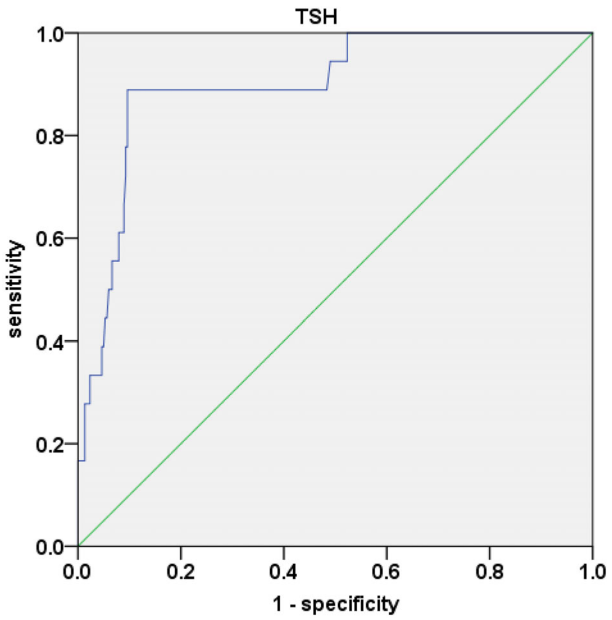

FIGURE 1 Receiver-operating characteristic curves of TSH to indicate RGBS. AUC = 0.899 (P < .001); 95% CI, 0.828 to 0.969; identified TSH cut-off value = 3.87 μIU/mL; Youden index = 0.793; sensitivity, 88.9%; specificity, 90.4%. AUC, area under curve; RGBS, recurrent Guillain-Barré syndrome; ROC, receiver operating characteristic; TSH, thyroid-stimulating hormone

3.4 | Analyses of ROC curves of thyroid parameters

ROC-curve analysis was used to verify the accuracy of TSH for predicting RGBS. The optimal cut-off point for TSH was 3.87 μIU/mL for predicting RGBS (Figure 1). At this cut-off value, the sensitivity was 88.9% and the specificity was 90.4%.

4 | DISCUSSION

In this study, RGBS patients were found to have significantly higher serum TSH levels than GBS patients. TSH was an independent risk factor for RGBS. Our ROC-curve analysis shows that the risk of recurrence in GBS patients increased when the TSH concentration was over 3.87 μIU/mL. One study showed that TSH levels were lower in GBS patients compared with healthy controls[12]; further large-scale studies are needed to investigate the effect of TSH on GBS and RGBS. Although our results do not show that FT3 was independently associated with RGBS, FT3 was lower in RGBS patients, prompting further thinking about the possible relationship between thyroid hormone (TH) and RGBS.

Extensive experiments have demonstrated that the process of myelination in multiple sclerosis patients depends on TH, which can enhance remyelination and relieve demyelination by shifting the demyelination/remyelination ratio toward remyelination and promoting the development of oligodendrocyte lineage cells.[13] Despite numerous studies that depicted the mechanism by which TH affects myelin in the central nervous system (CNS),[14,15] it remains to be determined whether TH could induce recurrences in patients with GBS by influencing peripheral nerve myelination.

A relationship has been demonstrated between TSH and a variety of neurological diseases, including multiple sclerosis, diabetic peripheral neuropathy, and Alzheimer disease.[16,17] Research has also shown that malondialdehyde (MDA), a final product of oxidative stress, is elevated in a higher TSH concentration group compared with a control group. This may indirectly suggest increased oxidative stress, an inadequate increase in antioxidant status, and changed lipid metabolism in the higher TSH group.[18] Some studies supported a strong positive correlation between oxidative stress and TSH levels, because of the increasing oxidative stress in patients with higher TSH.[19] Free radical toxicity due to unbalanced maintenance of cellular redox levels has been implicated in the pathogenesis of some CNS diseases, such as amyotrophic lateral sclerosis (ALS) and Alzheimer disease.[20] Furthermore, recent studies showed that increased reactive oxygen species (ROS) and imbalance of MDA and antioxidant activity in the plasma of GBS patients compared with healthy controls triggered activation of the mitogen-activated protein (MAP) kinase pathway in Schwann cells and caused subsequent axonal damage.[21,22] Excess ROS, as a product of oxidative stress, has been widely considered to be a mediator of demyelination and axonal injury.[21] Thus, it is possible that TSH induces oxidative damage through oxidative stress. Although some studies reported patients with a combination of GBS and hyperthyroidism or hypothyroidism,[23,24] the relationship between TSH and RGBS remains unclear. Our results demonstrate an association between TSH and RGBS, but it is unknown whether TSH can indirectly cause repeated myelin and axonal damage in RGBS patients through oxidative stress.

The frequency of recurrent GBS in this study was 6.0%, which is consistent with previous reports. Our study has also demonstrated a higher incidence of antecedent illness in RGBS patients, which is an independent risk factor for GBS recurrence. It is unclear whether recurrence is associated with repeated immune cross-reactivity triggered by antecedent illness as well as the immune status of host. We found that patients with GBS who were considered on the basis of nerve conduction studies to have AIDP were more likely to have recurrences, and that AIDP is an independent risk factor for RGBS. The underlying mechanism of more frequent recurrences of AIDP may be caused by defective Fas function, which is expressed by activated T lymphocytes and inhibits the immune response by mediating T-lymphocyte apoptosis to eliminate autoimmune T cells.[25]

Our study has some limitations. First, monophasic recurrent GBS in some patients may have occurred after the end of the study period. Second, although it is meaningful for us to suggest thyroid parameters as RGBS biomarkers, the mechanisms by which these biomarkers affect RGBS remain to be confirmed and investigated. Third, the number of RGBS patients in this study was small, and the impact of thyroid parameters on RGBS will need further large-scale studies.

Finally, TSH may have a role in determining which patients with GBS have a high risk of developing RGBS. GBS patients with AIDP and antecedent illness may be more likely to have a recurrence.

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CONFLICT OF INTEREST

None of the authors has any conflict of interest to disclose.

AUTHOR CONTRIBUTIONS

All authors assisted with writing, editing, and review of the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author.

ETHICAL PUBLICATION STATEMENT

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

ORCID

Tao Peng https://orcid.org/0000-0003-2473-997X

REFERENCES

1. Wakerley BR, Yuki N. Infectious and noninfectious triggers in Guillain-Barré syndrome. *Expert Rev Clin Immunol*. 2013;9:627-639.
2. Grand' Maison F, Feasby TE, Hahn AF, et al. Recurrent Guillain-Barré syndrome. Clinical and laboratory features. *Brain*. 1992;115:1093-1106.
3. Das A, Kalita J, Misra UK. Recurrent Guillain Barré syndrome. *Electromyogr Clin Neurophysiol*. 2004;44:95-102.
4. Ishii J, Yuki N, Kawamoto M, Yoshimura H, Kusunoki S, Kohara N. Recurrent Guillain-Barré syndrome, Miller Fisher syndrome and Bickerstaff brainstem encephalitis. *J Neurol Sci*. 2016;364:59-64.
5. Malek E, Salameh J. Guillain-Barré syndrome. *Semin Neurol*. 2019;39:589-595.
6. Tu Y, Gong X, Zeng G, et al. Differences in thyroid function and autoantibodies in the development of Guillain-Barré syndrome vs chronic inflammatory demyelinating polyradiculoneuropathy. *Front Neurol*. 2020;11:1018.
7. Van der Meché FG, Van Doorn PA, Meulstee J, et al. Diagnostic and classification criteria for the Guillain-Barré syndrome. *Eur Neurol*. 2001;45:133-139.
8. Wakerley BR, Uncini A, Yuki N. Guillain-Barré and Miller Fisher syndromes---new diagnostic classification. *Nat Rev Neurol*. 2014;10:537-444.
9. Ruts L, Drenthen J, Jacobs BC, van Doorn PA, on behalf of the Dutch GBS Study Group. Distinguishing acute-onset CIDP from fluctuating Guillain-Barré syndrome: a prospective study. *Neurology*. 2010;74:1680-1686.
10. Hughes RA, Newsom-Davis JM, Perkin GD, et al. Controlled trial prednisolone in acute polyneuropathy. *Lancet*. 1978;2:750-753.
11. Ho TW, Mishu B, Li CY, et al. Guillain-Barré syndrome in northern China. Relationship to *Campylobacter jejuni* infection and anti-glycolipid antibodies. *Brain*. 1995;118:597-605.
12. Huang Y, Ying Z, Chen Z, et al. Thyroid hormone level is associated with the frequency and severity of Guillain-Barré syndrome. *Int J Neurosci*. 2017;127:893-899.
13. Zhang M, Ma Z, Qin H, Yao Z. Thyroid hormone potentially benefits multiple sclerosis via facilitating remyelination. *Mol Neurobiol*. 2016;53:4406-4416.
14. Ausó E, Lavado-Autric R, Cuevas E, et al. A moderate and transient deficiency of maternal thyroid function at the beginning of fetal neocorticogenesis alters neuronal migration. *Endocrinology*. 2004;145:4037-4047.
15. Zhang M, Zhan XL, Ma ZY, Chen XS, Cai QY, Yao ZX. Thyroid hormone alleviates demyelination induced by cuprizone through its role in remyelination during the remission period. *Exp Biol Med (Maywood)*. 2015;240:1183-1196.
16. Long Y, Zheng Y, Chen M, et al. Serum thyroid-stimulating hormone and anti-thyroglobulin antibody are independently associated with lesions in spinal cord in central nervous system demyelinating diseases. *PLoS One*. 2014;9:e100672.
17. Zhao W, Zeng H, Zhang X, et al. A high thyroid stimulating hormone level is associated with diabetic peripheral neuropathy in type 2 diabetes patients. *Diabetes Res Clin Pract*. 2016;115:122-129.
18. Torun AN, Kulaksizoglu S, Kulaksizoglu M, Pamuk BO, Isbilen E, Tutuncu NB. Serum total antioxidant status and lipid peroxidation marker malondialdehyde levels in overt and subclinical hypothyroidism. *Clin Endocrinol*. 2009;70:469-474.
19. Cebeci E, Alibaz-Oner F, Usta M, Yurdakul S, Erguney M. Evaluation of oxidative stress, the activities of paraoxonase and arylesterase in patients with subclinical hypothyroidism. *J Invest Med*. 2012;60:23-28.
20. Dib M, Garrel C, Favier A, Robin V, Desnuelle C. Can malondialdehyde be used as a biological marker of progression in neurodegenerative disease. *J Neurol*. 2002;249:367-374.
21. Smith KJ, Kapoor R, Felts PA. Demyelination: the role of reactive oxygen and nitrogen species. *Brain Pathol*. 1999;9:69-92.
22. Rodella U, Scorzeto M, Duregotti E, et al. An animal model of Miller Fisher syndrome: mitochondrial hydrogen peroxide is produced by the autoimmune attack of nerve terminals and activates Schwann cells. *Neurobiol Dis*. 2016;96:95-104.
23. Pandolfi C, Filippi C. Guillain-Barré syndrome associated with hypothyroidism. Report of a case. *Minerva Med*. 1989;80:1129-1131.
24. Ali A, Malik WT, Raja U, Khan FA, Javaid I. Thyroid storm, a mimicker of Guillain-Barre syndrome: a case report. *Cureus*. 2020;12:e9918.
25. Comi C. Fas-mediated T-cell apoptosis in chronic inflammatory demyelinating polyneuropathy. *J Peripher Nerv Syst*. 2011;16(Suppl 1):45-47.

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